CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 05



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Recap of Previous Lecture



Topic — Architecture Type (Based on Size of Input)

Topic — Microoperations

Topics to be Covered



Topic Microoperations

Topic Micro-programming

[MCQ]

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1, (3000)	R1 ← M[3000]	2
LOOP:	MOV R2, (R3)	R2 ← M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M [R3] ← R2	1
	INC R3	R3 ← R3 + 1	1
	DEC R1	R1 ← R1 – 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. The number of memory reference for accessing the data in executing the program completely is

A 10

B 11

C 20

D 21

[MCQ]

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1, (3000	R1 ← M[3000]	2
LOOP:	MOV R2, (R3)	R2 ← M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M [R3] ← R2	1
	INC R3	R3 ← R3 + 1	1
	DEC R1	R1 ← R1 – 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

A 100

B 101

C 102

D 110

[MCQ]

#Q. Consider the following program segment. Here R1, R2 and R3 are the general-purpose registers.

address

	Instruction	Operation	Instruction Size (no. of words)
1000	MOV R1, (3000)	R1 ← M[3000]	2 = 8 bytes
1008 LOOP:	MOV R2, (R3)	R2 ← M[R3]	1 = 4 bytes
1012	ADD R2,R1	R2 ← R1 + R2	1
1016	MOV (R3),R2	M [R3] ← R2	1
1020	INC R3	R3 ← R3 + 1	1
1024	DEC R1	R1 ← R1 – 1	1
1028	BNZ LOOP	Branch on not zero	2
1036	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

4 bytes

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on the stack?

A 1005

B 1020

C ✓1024

D 1040

Note :-

If any interrupt occurs during execution of an instruction, then CPU Completes execution of current instructions, and stores address of next instruction onto the stack.

1 word = 32 bits = 4 bytes

Ques) In prev. quest^{n} if mem. is word addressable with word size 4 bytes, then what return add. will be pushed on to the stack ?

		size in words
1000	MOV R1, (3000)	2
1002	MOV R2, (R3)	1
1003	ADD R2, R1	1
1004	MOV (R3), R2	1
1005	INC R3	1
1006	DEC R1	1
1007	BNZ loop	2
1009	HALT	1

Ans = 1006

[NAT]

PYQ GATE- 2021

#Q. Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY [X] denotes the content at the memory location X.

	Instruction	Semantics	Instruction Size (bytes)
1000	MOV R1, (5000)	R1 ← MEMORY[5000]	4
1004	MOV R2, (R3)	R2 ← MEMORY[R3]	4
1008	ADD R2, R1	R2 ← R1 + R2	2
1010	MOV (R3), R2	MEMORY[R3] ← R2	4
1014	INC R3	R3 ← R3 + 1	2
1016	DEC R1	R1 ← R1 – 1	2
1018	BNZ 1004	Branch if not zero to the given absolute address	2
1020	HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3010 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is 50.

[NAT]

#Q. Consider a memory with size 128 bytes. The address size to access memory, if memory is byte addressable, is ____7____ bits?

$$\text{no. of cells} = \frac{128\,B}{1\,B} = 128 = 2^7$$

$\Downarrow$

address = 7 bits

[NAT]

#Q. Consider a memory with size 2048 bytes. The address size to access memory, if memory is word addressable(1 word = 4 bytes), is 9 bits?

$$\text{no. of cells} = \frac{2048\,B}{4B} = \frac{2^{11}}{2^2} = 2^9 \Rightarrow \text{add.} = 9 \text{ bits}$$

[NAT]

#Q. Consider a memory with size 16Mbytes. The address size to access memory, if memory is byte addressable, is 24 bits?

$$\text{no. of cells} = \frac{16\,MB}{1\,B} = 16M = 2^4 \cdot 2^{20} = 2^{24}$$

$\Downarrow$

add. = 24 bits

[NAT]

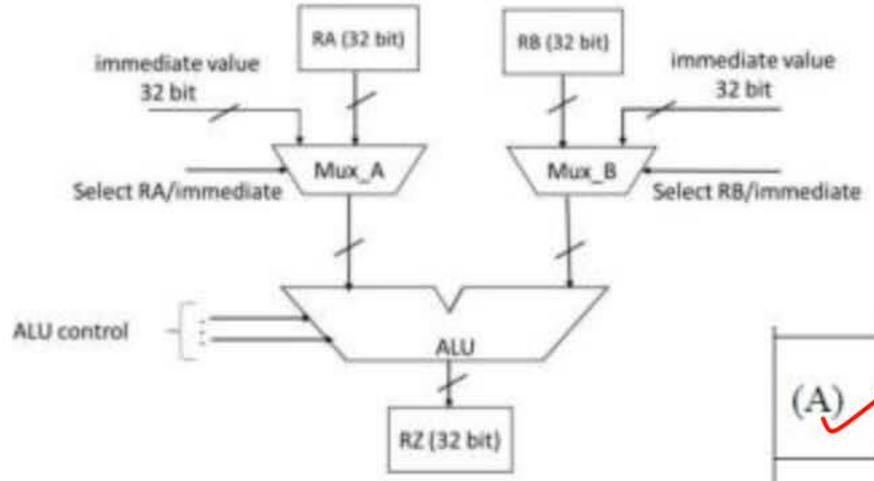
#Q. Consider a memory with size 64Kbytes. The address size to access memory, if memory is word addressable(1 word = 2 bytes), is 15 bits?

$$\text{no. of cells} = \frac{64KB}{2B} = 32k = 2^5 \cdot 2^{10} = 2^{15}$$

address = 15 bits

[MSQ] GATE-2025

Q.27	A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?



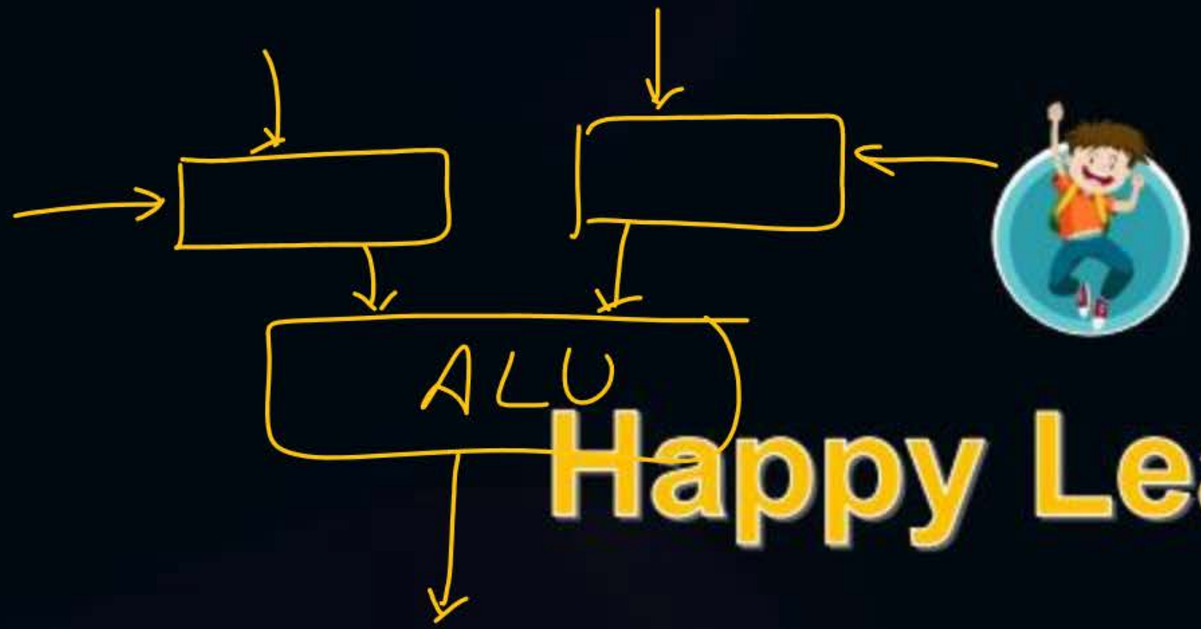
(A)	The data path can implement arithmetic operations involving two registers.
(B)	The data path can implement arithmetic operations involving one register and one immediate value.
(C)	The data path can implement arithmetic operations involving two immediate values.
(D)	The data path can only implement arithmetic operations involving one register and one immediate value.



2 mins Summary

Topic Microoperations

Topic Micro-programming



Happy Learning

THANK - YOU